

Karthik Rajgopal

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EDUCATION

University of California, Berkeley

MEng Mechanical Engineering (Robotics & Control Systems track) [GPA: 3.58/4.0]

Aug 2024 - May 2025

California, US

Courses: Experiential Advanced Control Design I (Model Predictive Control) & II (State Estimation), Modeling and Control of Multi-Agent Systems, Deep Tech Commercialization Strategies, Lean Launchpad, Product Development

Birla Institute of Technology and Science, Pilani - Pilani Campus

Bachelor of Engineering, Mechanical Engineering [GPA: 8.79/10]

Aug 2019 – May 2023

Rajasthan, India

Courses: Control Systems, Mechanisms & Machines, Computer-Aided Design, Advanced Manufacturing Process

SKILLS

Programming: Python, C, C++, MATLAB

CAD & FEA:

SolidWorks, Fusion 360, AutoCAD, Ansys

Robotics: ROS, Gazebo, PyBullet, Simulink

Others:

Arduino, 3D Printing, Figma

WORK EXPERIENCE

Project Assistant

Robert Bosch Centre for Cyber-Physical Systems, Indian Institute of Science

May 2023 – June 2024

Bengaluru, India

– **Collision Cone Control Barrier Functions (C3BF)** [Publication] | ROS, Python

- Developed a real-time safety filter that combines the concepts of collision cone and control barrier functions (CBF) with mathematical guarantees that can be implemented on top of a reference controller.
- Experimentally verified C3BF-QP-based controller on unicycle (Turtlebot) and bicycle (Ackermann) robot platforms.

– **Mechanical Design of Quadraped** | SolidWorks, Ansys, PyBullet

- Designed a 3-DOF robotic leg with compound planetary actuators for 25 kg payloads on challenging terrains.
- Applied topology optimization to the lower leg (shank), achieving 12% weight reduction and 18% strength increase; and validated performance using static structural analysis using load data from MPC-driven PyBullet simulations.
- Optimized the knee joint using roller chain and bearing calculations; coordinated vendor-based fabrication of EN8 steel sprockets via wire-cutting, and ensured CNC and metal 3D printed components met DFM and structural requirements.

PROJECTS

Learning-to-win Drone Racing (MEng Capstone Project) | Python

Mentor: Prof. Claire Tomlin, Professor, UC Berkeley

Sep 2024 – Present

California, US

- Designing drone racing controllers using deep reinforcement learning and game theory for competitive multi-agent scenarios.
- Leveraged reachability analysis using max-min Deep Deterministic Policy Gradient (DDPG) for safety-critical interactions.
- Tested Multi-Agent Proximal Policy Optimization (MAPPO) for high-level decision-making against adversarial agents.

HyperVaal: E-Bike Conversion Kit for Urban Commuters | SolidWorks, Ansys

Mentor: Dr. Ala Moradian, UC Berkeley

Jan 2025 – May 2025

California, US

- Designed and prototyped an e-bike conversion kit with a 750W BLDC motor, 60:1 drivetrain, and an adjustable pipe-clamp mounting system; integrated motors and gearboxes into a sealed powertrain using laser-cut steel and 3D printed PLA.
- Validated drivetrain performance through torque-speed modeling and multi-stage efficiency analysis; achieved target specifications of ≥ 2.8 Nm output torque and 89% system efficiency while ensuring cost-effective assembly and modularity.

Coverage Control for Aerial-Ground Robot Teams in Emergency Response | Python, SciPy

Mentor: Prof. Negar Mehr, Assistant Professor, UC Berkeley

Sep 2024 – Dec 2024

California, US

- Designed a hybrid Voronoi coverage controller for aerial-ground robot teams to reduce sensing delays in emergency response.
- Built 2D and 3D Python simulations using SciPy to evaluate bounded-region coverage and centroid-seeking performance.
- Demonstrated 88% reduction in sensor loss time via airdrop-based deployment and velocity-aware coordination control.

Optimized Path Planning for Parallel Parking | Python

Mentor: Dr. Charlott S. Vallon, UC Berkeley

Sep 2024 – Dec 2024

California, US

- Implemented a Model Predictive Control algorithm for precise and adaptive path planning in parallel parking.
- Tuned cost functions and algorithm parameters to ensure constraint satisfaction, smooth trajectories, and robust control.
- Validated the controller in simulation, achieving <5 cm final parking error with accurate and reliable trajectory tracking.

Design and RL-based Control of a Biped [Publication] | SolidWorks, MuJoCo

Mentor: Dr. Shashir NY Kolathaya, Assistant Professor, Indian Institute of Science

Aug 2022 – April 2023

Bengaluru, India

- Designed a low-cost, 3D printed modular 3-DOF-per-leg biped using BLDC-driven belt-pulley actuators (9:1 reduction).
- Led mechanical design, performed FDM 3D printing, and coordinated laser-cut aluminum fabrication with vendor.
- Assisted implementation of linear policy-based controller, tuning PD gains and validating locomotion stability on hardware.

COMPETITIONS

Runners Up - Berkeley Venture Capital Investment Competition (VCIC)

Dec 2024 - Feb 2025

- Conducted founder due diligence, valuation, and competitive analysis to assess venture investment opportunities.
- Participated in partner meetings, drafted term sheets with key terms, and performed market research on startups.